

Sadhvi Sharma

Senior AI Engineer | Agentic AI, LLM Applications and AI Platform Engineering

Calgary, Alberta, Canada | Open to Toronto and hybrid | sadhvisharma763@gmail.com | linkedin.com/in/sadhvi-sharma-5789a6249

Portfolio: sadv11.github.io | GitHub: github.com/sadv11

Permanent Resident of Canada. No sponsorship required. Available immediately.

PROFESSIONAL SUMMARY

Engineer building **agentic AI systems and the platforms that run them**, with five years across cloud infrastructure and telecom. I ship LLM applications where the safety controls are enforced in code rather than requested in a prompt: **Model Context Protocol tool servers**, anti-hallucination controls that refuse rather than guess, **structured output validated against strict schemas**, and evaluation suites that attack the system with prompt injection. Backed by production DevOps depth: CI/CD, containerisation, infrastructure automation and observability. Two and a half years on production on-call at Nokia against a contractual 99.9 per cent SLA gives me the operational side most AI work is missing.

CORE SKILLS FOR THIS ROLE

Agentic AI and LLM applications: agent orchestration, tool and function calling, multi-step workflows, conversation memory and **state management** with TTL-backed persistence, human-in-the-loop escalation, prompt engineering, structured output frameworks.

Open agent interfaces: Model Context Protocol (MCP) tool servers — built, published, and verified end to end against a live MCP client. Tool descriptions engineered as the model-facing contract.

Retrieval and grounding: retrieval-augmented generation, embeddings (Amazon Titan V2, sentence-transformers), **vector search over pgvector**, cosine similarity thresholds, chunking and context management, source attribution in responses.

AI governance, evaluation and safety: anti-hallucination controls with explicit confidence floors and refusal paths; **explainability and traceability** — every verdict and refusal reports the evidence and threshold behind it; evaluation suites with held-out question sets, graded cases and edge-case validation; **prompt-injection guardrails**; policy enforced in code rather than in the system prompt.

AI platform and cloud: Amazon Bedrock, AWS AgentCore runtime, SageMaker, Lambda, ECS, EKS, DynamoDB, S3, API Gateway. Microsoft Azure: Container Apps, AKS, Entra ID, Key Vault, Log Analytics.

Productionisation and DevOps: CI/CD (GitHub Actions, Azure DevOps), Docker, Kubernetes, Terraform, CloudFormation, Bicep, Helm, Argo CD. Secrets management, OIDC federation with no stored credentials, least-privilege IAM.

Observability and reliability: Prometheus, Grafana, CloudWatch, structured logging, service level objectives and error budgets, blameless postmortems, production on-call.

Regulated environments: control mapping for SOC 2, PCI DSS, PIPEDA, provincial FIPPA and OSFI B-13; enforced data residency; privacy-aware architecture.

Languages and APIs: Python (boto3, FastAPI, Flask), Bash, HCL, SQL. REST API design; **distributed and event-driven systems** on SQS, Kinesis and Lambda, and a twelve-service distributed application on Kubernetes.

SELECTED AI ENGINEERING WORK

All public and independently verifiable at github.com/sadv11. Every repository passes automated tests on every change.

MCP tool server with policy enforced in code - terraform-guard-mcp

A **Model Context Protocol server** exposing five tools to any MCP client, verified end to end against a live client: full handshake, tool discovery, and invocation. Its decision tool returns a verdict computed in Python, not by the model, so **no wording in the input can argue it down** — including a test case whose payload is named to look pre-approved and contains an instruction to ignore prior instructions. **34 tests**; the pipeline removes each of three guards on purpose and fails if the suite still passes. **Then I attacked my own tool and found a prompt-injection weakness in it:** attacker-controlled text was printed into the model-facing output, able to forge a false system boundary. Fixed at parse time so all five tools are covered, with tests asserting structure rather than vocabulary.

Anti-hallucination and evaluation - bedrock-agentcore-customer-agent

A customer support agent on **Amazon Bedrock AgentCore**. Retrieval returns the best match **and its similarity score**; below an explicit floor the agent returns `NO_CONFIDENT_MATCH` with the score and threshold and escalates to a human rather than answering from a weak match. Evaluation runs against **40 held-out questions** scoring both correct answers and correct refusals. Found that the confidence floor is model-specific and had to be re-measured when the embedding model changed — carrying the old value across would have refused roughly half the answerable questions.

Retrieval-augmented generation on Bedrock - bedrock-rag-app

Document question answering: extraction, chunking with overlap, **Titan Embeddings V2 at 1024 dimensions**, **vector search over pgvector**, and grounded generation that reports missing context instead of inventing an answer. Deployed and live. Data residency limitations documented explicitly rather than glossed over.

Structured output and guardrails - structured-test-agent, smart-ai-agent, aws-ec2-read-only-agent

Forced tool choice against a strict JSON Schema so responses are machine-consumable by construction, with tests asserting the schema stays strict rather than advisory. An agent with tool use and persistent vector memory, graded by an evaluation suite that **attacks it with prompt injection and SQL injection**. And an AI assistant over a cloud account that is **prevented in code from calling any mutating operation**, via an allow-list checked at the client layer before a request is ever signed — a persuaded model still cannot act.

WORK EXPERIENCE

AI Hardware and Technology Specialist

Meta, through Influence Marketing | Calgary, Alberta | October 2025 to present

- Technical enablement on Meta AI products and extended reality hardware. Delivered training to more than 50 staff, reducing escalated issues by approximately 30 per cent.
- Translate AI capabilities and limitations for non-technical audiences — the stakeholder-facing half of AI delivery.
- All AI and cloud engineering above built and maintained outside working hours.

Solution Engineer, Cloud Core Network

Nokia | Delhi, India | December 2022 to July 2025

- Operated 5G mobile core systems as containerised workloads on Kubernetes and OpenStack in live production for **Bell Canada, T-Mobile US and Orange**, against a contractual **99.9 per cent** availability guarantee.
- Delivered a staged multi-site 4G to 5G upgrade with **zero subscriber downtime**. Nokia appreciation award.
- Production on-call, root cause analysis, and zero-downtime rolling upgrades gated by health checks — the operational discipline that production AI platforms require.
- Worked directly with international carrier customers in a highly regulated, contractually governed environment.

RF and Systems Engineer

AA Electro Magnetic Test Labs | India | 2018 to 2020. Electromagnetic compatibility testing against published standards.

EDUCATION

Master of Technology, Electronics and Communication Engineering

University of Delhi, India. Full merit scholarship.

Bachelor of Technology

Shri Mata Vaishno Devi University, India.

CERTIFICATIONS AND CONTRIBUTIONS

- Microsoft Certified: Azure Fundamentals (AZ-900). AWS Certified Solutions Architect - Associate, scheduled September 2026.
- aquasecurity/trivy-checks pull request 602 — fix allowing TLS 1.3 in a security policy that incorrectly rejected it. Open, awaiting maintainer review.

VOLUNTEER EXPERIENCE

Area Director, Toastmasters International (District 42). Accountable for four to six member clubs, none reporting to me, achieving results through persuasion rather than authority. Second place at Division level, International Speech Contest.